

Recovering word-prosody in an extinct language family

The case for stress accent in Luwic

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Typology in Historical Phonology

Uppsala

25.08.2026



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- Luwic languages = Anatolian, Indo-European, and extinct

What was the word-prosody of the Luwic languages and Proto-Luwic like?

- Prototype definition of **stress-accent** given by Hyman (2009)
- Set of 8 (some phonetic) features

Stress accent	Pitch accent	Tone
English	W. Basque	Tokyo Japanese
	Luganda	Mandarin

Premise

The more **SA-like Luwic** is, the more features of the **prototype** set it will exhibit
(and *vice versa*)

Language	Subcorpus	Date (ca.)	Tokens (ca.)	Writing system
Luwian	Cuneiform	1600–1200 BCE	14,000	Logosyllabic (greater phonetic detail)
	Hieroglyphic	1500–700 BCE	14,500	Logosyllabic (lower phonetic detail)
Lycian	Lycian A	500–300 BCE	6,000	Alphabetic
	Lycian B			
Lydian		700–100 BCE	1,500	Alphabetic
Carian		600–200 BCE	600	Alphabetic
Sidetic		450–200 BCE	< 600	Alphabetic
Pisidian		100–300 CE	< 600	Alphabetic

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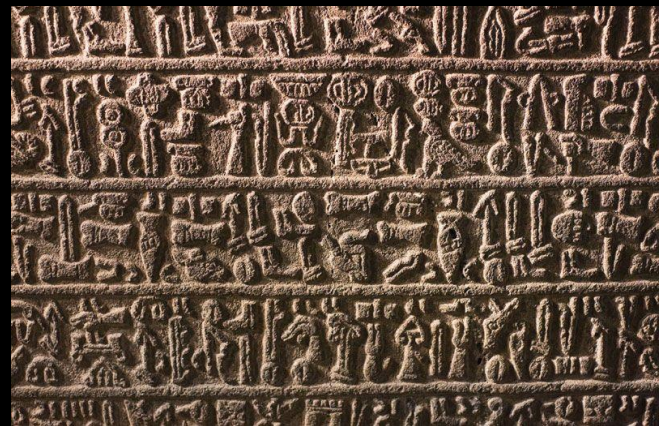
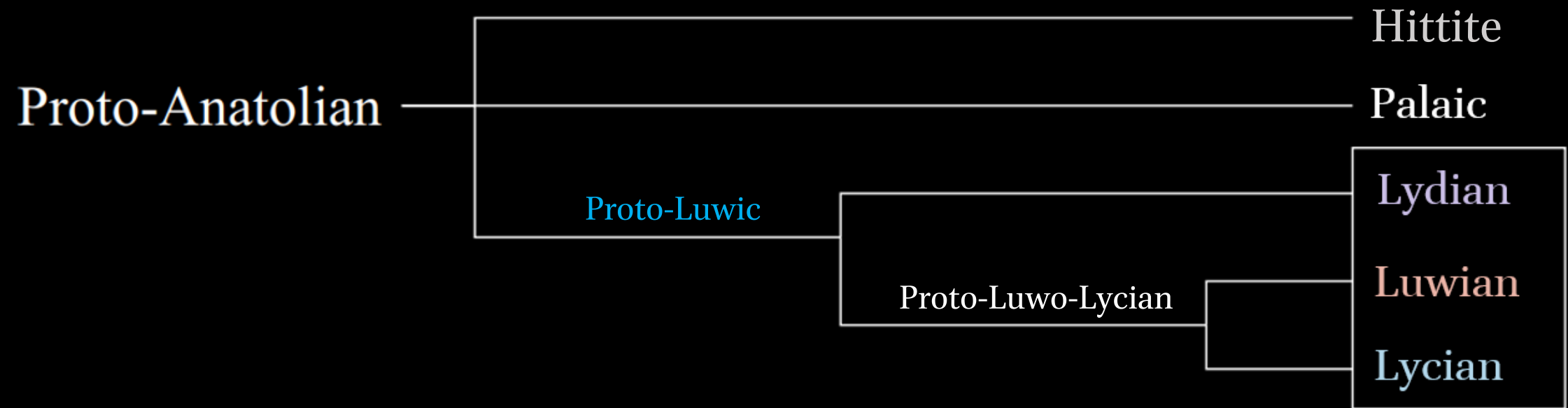
Lydian

Fig. 1

Lycian

ἄβ ἄε: μάσει ψνε: ὠσσω ἱεῖρα:
 ὠεταρρα + τε δ ἄε με: μοτ ἄ +:
 πρεν ἄε ἰα + ε: πρεν ἄρ ρ τ ἄ τε:
 ἱτρ τψ: ρ τ ἄε: ἄ βε: σ ἄε ἰ ἄ ἄ: + ρ ρ γε
 τ ψ τε: τε κ ἄ: κ βε: + ρ ρ ρ εἰ ἄ με:
 τ ψ  ε: ἄ ἄ ἄ: κ ρ στ τε: ἄ ρ ἄε:
 ρ ἄ ρ ψνε: σ ἄε τ ἄ + ε: τ ρ χ με ἄ
 + ο ρ ἄ ρ ε: ὠσσω ἱεῖρα ἄ: + β ἄ ρ ο σ ἄ
 ἱ α σ ψ νε: ὠ ρ ἄ ρ εν ρ: ο + ρ ἱ ρ τ ρ: κ ο μ ἄ ἱ ε ἄ ἄ



OBLIGATORINESS: every lexical word has **at least** one syllable marked for the highest degree of metrical prominence (primary stress)

CULMINATIVITY: every lexical word has **at most** one syllable marked for the highest degree of metrical prominence

(Hayes 1995: 24f.; Hyman 2006: 231)

Lydian: Both **O** & **C** manifestly satisfied under accent analysis by Eichner (1986; 1987)

Lycian: Widespread syncopation suggests **C** and probably **O**

Luwian: Both **O** & **C** **difficult to demonstrate** synchronically

Prehistory: **O** & **C** satisfied by Hittite and **C** is routinely reconstructed for Proto-Indo-European (cf. Yates 2017; Byrd 2018: 2069f.)

- A. **Stress** location **not** reducible to first/last syllable (non-demarcative)
- B. **Stressed** syllables show positional **prominence** effects
 - i. More consonant-, vowel-, and tone **oppositions** on stressed syllables
 - ii. Segment **strengthening** in stressed syllables
- C. **Unstressed** syllables show positional **non-prominence** effects
 - i. Less consonant-, vowel-, and tone **oppositions** on **unstressed** syllables
 - ii. Segment **weakening** in **unstressed** syllables
- D. **Stress** shows **cyclic** effects
- E. **Stress** shows **rhythmic** effects lexically/postlexically
- F. Lexical **stresses** interact at the **postlexical** level, e.g. **compounding/phrasal stress**
- G. Lexical **stress** provides the designated terminal elements for the assignment of intonational tones
- H. “Other arguments that every syllable is in a metrical constituent which can be globally referenced”

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- D. Stress shows **cyclic** effects
- ~~E. — Stress shows **rhythmic** effects lexically/postlexically (?) (cf. Excursus)~~
- F. Lexical stresses interact at the postlexical level, e.g. **compounding/phrasal stress**
- ~~G. — Lexical stress provides the designated terminal elements for the assignment of intonational tones~~
- ~~H. — “Other arguments that every syllable is in a metrical constituent which can be globally referenced”~~

B. Accented syllables show prominence effects

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Luwian

Cuneiform **plene** spelling <(CV-)V(-VC)> Primarily denotes **length**
(cf. Kloekhorst 2014: 13ff.; Rieken 2017: 19f.; Yakubovich & Rieken 2026)

Frequent correlate of **inherited accented** short vowels
(Melchert 1994: 261ff.; Yakubovich & Rieken 2026:101f.)

Cf. e.g. *a-aš-ti* [á:sti] 'is' < PIE **h₁és-ti* (cf. Ved. *ásti*)

za-a-aš [tsá:s] 'this' < PIE **kós* (cf. Hitt. *kās*)

Strongly suggests lengthening of **accented** syllables

Lycian

No direct evidence of prominence effects

B. Accented syllables show prominence effects

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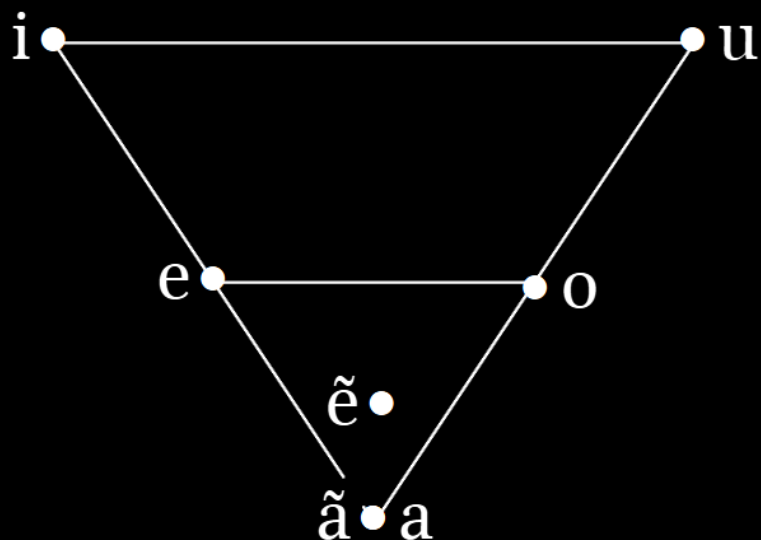
Lydian

More phonological vowel oppositions in accented syllables (Eichner 1986a; 1987)

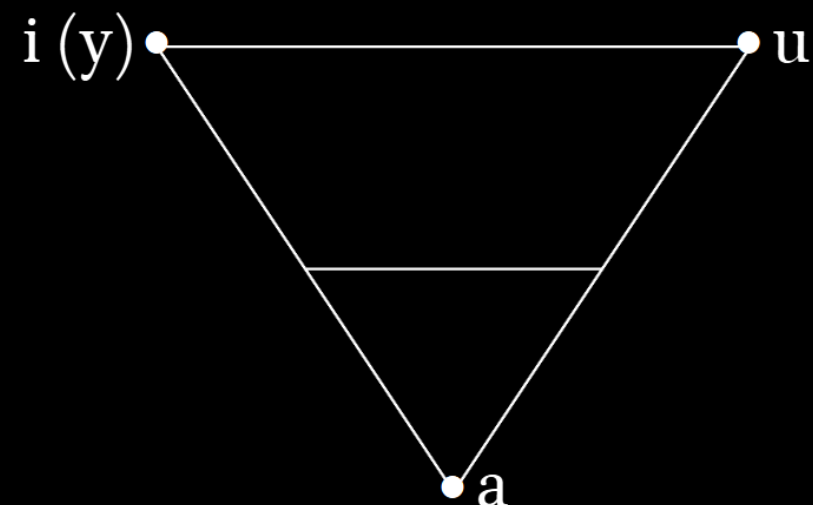
Vowel signs permitted in **accented** syllables: *a e i o u ã ë*

Vowel signs permitted in **unaccented** syllables: *a i (y) u*

Accented vowel space



Unaccented vowel space



B. Accented syllables show prominence effects

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Lydian

Vowel **lengthening** in **accented** position

(West 1972: 167; Eichner 1986b: 215f.; Melchert 1994: 343; Gérard 2005: 37f.; but cf. Kloekhorst 2023: 118f.; 120)

Indicated by optional digraph *aa* (and possibly *ii*)

Cf. e.g. <i>taadas</i> (LW 14.6) ‘father’	[tá:ðas]	/táðas/
<i>qiraad</i> (LW 13.3) ‘property’	[kʷirá:ð]	/kʷiráð/

Carian

Vowel denoted by letter *o* (< Proto-Luwic **ā*, ? *Ūw*) only in **accented** position (?)

(Adiego 2019: 19f.; 23)

Cf. e.g. *mno*- ‘boy (vel sim.)’
qmoλ- ‘priest’

B. Accented syllables show prominence effects

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Luwo-Lycian (or Proto-Luwic) prehistory

Čop's law

$*L > F / \acute{e}_-$ (*L*enis, *F*ortis)

Originally formulated for **Luwian** by Čop (1970)

mallit- 'honey' < PIE **mélit*-

maddu- 'wine' < PIE **méd^hu*-

Also proven for **Lycian** by Kloekhorst (2012: 261)

ebette DAT/LOC.PL pronoun < PIE **h₁ob^héd^hos* (cf. Hitt. *apēdas*)

Lydian status = **unsure!**

C. Unaccented syllables show non-prominence effects

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Luwian (prehistory)

(?) “Reduction” of pretonic $*e > /i/$ ($*é > /a/$)

Cf. e.g. HLuw. *tipas-* ‘sky’ < tr. PIE $*neb^h-és-$

(Melchert 1994: 240; Hajnal 1995: 63)

Lycian

Widespread syncope (cf. Hajnal 1995)

Cf. e.g. *tédi-* ‘father’ → $*tédehe/i-$ > $te\vartheta\vartheta e/i-$ ‘paternal’

iterative suffix PIE $*-ské/ó-$ > $*'-se-$ > $-s-$

PIE $*g^{wh}n-ské-ti$ > $*qánseti$ > *qástti* ‘destroys’

(cf. Serangeli 2018; Billing 2025: 264ff.)

Lydian

Widespread syncope & apocope

(Melchert 1994: 373; Sasseville 2017: 137¹³)

Cf. e.g. $*mršalá-$ > *mršla-* ‘defiler’

(Billing 2025a: 373)

3SG.PRES.ACT $*-ti$ > $-t$

Carian

Widespread syncope (Simon 2023)

Luwian

Morphologically conditioned accent in some categories

Cf. non-leniting *a*-stem vs. leniting *ā(i)*-stem verbs (Billing 2025a: 316ff.)

Non-leniting class = frequent **non-plene** spelling of stem-final vowel + **plene** elsewhere

Leniting class = frequent **plene** spelling on stem-final vowel

	<u>Barytone non-leniting class</u>	vs.	<u>Oxytone leniting class</u>
3PL.PRES/PRET.ACT	<i>li-i-la-an-ti</i> ‘pacify’ (KUB 46.38 ii 24’) [lí:la:nti]	vs.	<i>za-ar-pa-a-in-ta</i> ‘put on shoes’ (KUB 35.107 iii 23’) [tsarpá:inta]
2SG.IMPV.ACT	<i>ha-a-ši-ia</i> ‘satisfy!’ (KBo 23.50 iii 22’) [χá:sija:]	vs.	<i>a-ru-na-a</i> ‘raise!’ (KUB 35.132 ii 6’) [aru(:?)ná:]
same deriv. base	€€ <i>pa-a-ši-ḥa-aḥ-ḥa</i> (KUB 40.1 obv. 9) [pá:siχa:χa]	vs.	€€ <i>pa-ši-ḥa-a-ti</i> ‘rub’ (KUB 14.3 ii 25) [pasixá:ti]

Lycian

Cf. disyllabic *a*-stem noun pairs like:

A *lãta*- [láⁿta-] ‘dead person’ vs. A *xñta*- [kɲtá-] ‘trust (vel sim.)’

Lydian

Cf. disyllabic *a*-stem noun pairs like:

NOM.SG *taadas* ‘father’ (LW_{14.16}) vs. *qašaas* ‘fee, rental price’ (LW_{10.13})
[tá:ðas] [kʷaɕás]

Proto-Luwic

Cf. e.g. **tódi*- ‘father’ > Lyc. A *tedi*- ‘id.’ ≈ CLuw. *tāti*- [táxti-] ‘id.’

**sirmǎ*- ‘temenos’ > Lyc. A *hrm̃má*- ‘id.’ ≈ Lyd. *širmá*- ‘id.’

Luwian (Iṣtanuwa dialect)

Possible case of secondary accent: *dussaniyalla-* ‘creator of strong ones (vel sim?)’

Accentuation *dússaniyálla*s assumed on metrical grounds (Billing 2025b)

**dússan-* ‘strong one’ (< tr. PIE **téuH-s-on-*)

→ **dussani-* ‘to make strong ones’ (Sasseville 2021: 148–151)

→ *dussaniyálla-* ‘creator of strong ones’ (deriv. suffix accent -*álla-*)

↓
→ *dùssaniyálla-* (secondary accent from derivational base **dússan-*)

Cf. Yates 2025 for a similar analysis of Hittite

Lycian and Lydian: **no positive evidence of cyclic stress effects (?)**

Luwian

Evidence for differential accentuation **adverb** vs. **preposition** (Melchert 2020: 265; 2024: 15)

annan 'below'

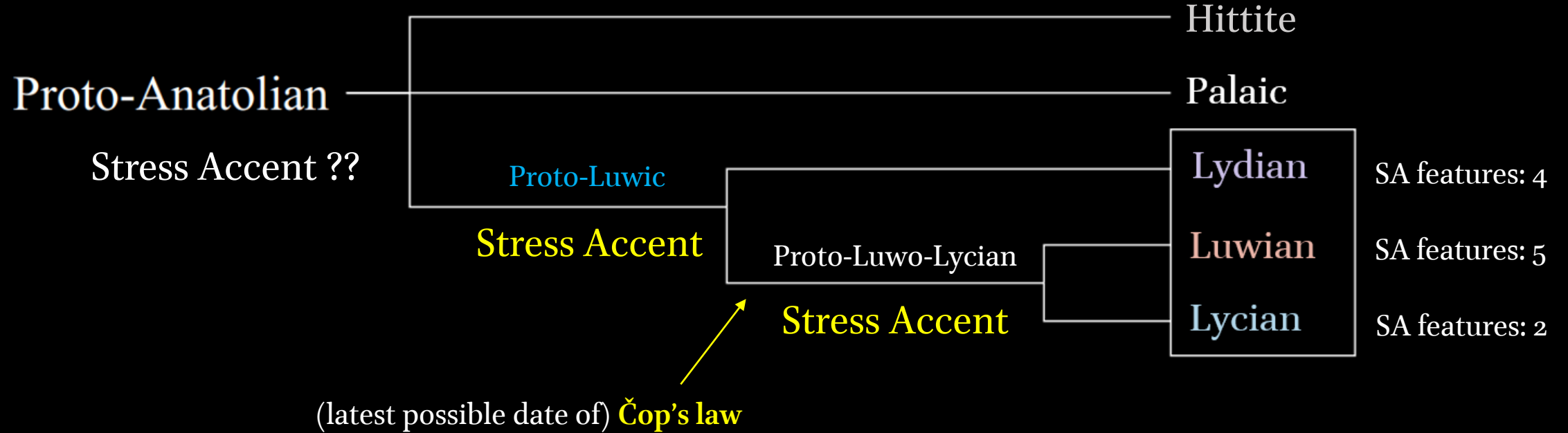
Adverb	<i>a-an-na-an</i>	<i>za-ar-pí-in-za</i>	<i>za-ar-pa-a-in-ta</i>	(KUB 35.107 iii 23')
	below.ADV	shoe.ACC.PL	to.shoe.3PL.PRET.ACT	
	'below, they put on shoes'			

vs.

Preposition	<i>an-na-a-an</i>	<i>pa-a-ta-an-za</i>	<i>du-ú-wa-an-du</i>	(KUB 9.6 iii 29')
	below.PREP	foot.DAT/LOC.PL	to.place.3PL.IMPV.ACT	
	'may they place (him) below the feet'			

a-an-na-an = [á:n:an] vs. *an-na-a-an* [an:á:n]

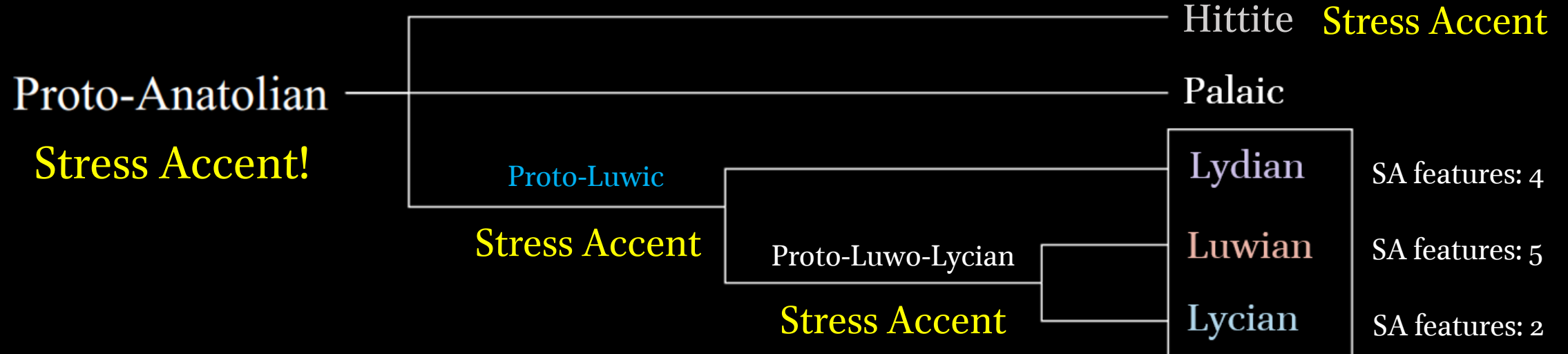
Criterion	Luwian	Lycian	Lydian
Accent location not demarcative	Y	Y	Y
Accented syllables show prominence effects	Y	N	Y
Unaccented syllables show non-prominence effects	Y ?	Y	Y
Accent shows cyclic effects	Y ?	?	?
Compounding/phrasal accent	Y ?	?	?



Hittite already analysed as **SA** by Kloekhorst (2014) and Yates (2017)

Prototypical **SA** features in Hittite:

1. **CULMINATIVITY** and **OBLIGATORINESS** satisfied (Yates 2017: 108)
2. **Weakening of unaccented** inherited **e* to /a/ and /i/
e.g. weak *i*-stem suffix *-ai-* < PIE **'-ei-*
nēpis- 'sky' < **néb^h-es-*
3. **Apocope of inherited unaccented** **e* in final position (Pisani 1952: 322)
e.g. *takku* /tak^w:/ 'if' < PIE **to-k^we*
4. **Shortening of inherited unaccented long vowels**
e.g. *tēkan* 'earth' < PIE **d^héǵ^h-ōm*
5. **Lengthening of accented** vowels
e.g. Hitt. /á:/ < PIE **ó*



1. Each Luwic language exhibits at least one prototypical SA feature
2. Most prototypical SA features are found in at least one Luwic language
3. Every Luwic language more or less fits a SA profile
4. Proto-Luwic should be reconstructed with a SA profile
5. Since Hittite likely has SA, Proto-Anatolian probably also did

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Luwian metre

(Billing 2025b; cf. also Watkins 1986: 61; 1995: 146–148, Schürr 2003: 110; Eichner 1993: 109–110)

- Syllable- and stress-counting
- Consists of **trochees** | ' ˘ | and **dactyls** | ' ˘ ˘ |

The Iṣtanuwan “*dussaniyalla*-song”

dúššani|yállaš=mi | *áyattar | páyu ||
(a)ntí=mi=tta | *dúwanta | wášu | páddu ||

' ˘ ˘ | ' ˘ ˘ | ' ˘ ˘ | ' ˘ ||
' ˘ ˘ | ' ˘ ˘ | ' ˘ | ' ˘ ||

- Syllable- and stress-counting
- Primarily consists of anapaests | ˘ ˘ |

wāntas wic | *n areλ* | *katarēd=k=* | *ms iškos* ||
 ‘ , ‘ | ‘ ‘ | ‘ ‘ | ‘ ‘ ||

 Red bar under *wā*, Blue bar under *nan*, Red bar under *tas*, Blue bar under *are*, Blue bar under *red*, Red bar under *k=*, Blue bar under *ms*, Blue bar under *iš*, Red bar under *kos*.

Lycian metre on poetic Lycian B inscriptions TL 44c-d and TL 55= **unsolved!**

Hyman's criterion E: “Stress shows rhythmic effects lexically/postlexically”

The stress-counting metres of the Luwic languages are indicative of stress-timed languages, exhibiting the rhythmic properties associated with SA

Hittite metre is similar, but does not count syllables